# PROJECT FRAMING DOCUMENT — AI AUDIT MINI-PROJECT

**National Polytechnic School of Oran Maurice Audin (ENPO-MA), Algeria**

Information Systems Engineering and Management

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**Purpose and academic status.** This document defines the scope, methodology, and organization of the AI audit mini-project required for Auditing AI Systems of the Information Systems Audit course. It serves as the contractual framing between the team and the instructor before the audit work begins.

**Deliverables:** audit report, reproducible notebook, corrective recommendations.

## 1. Context and Audit Objective

The audited model is a binary classifier trained on the public *Adult Income* dataset (UCI). Its task is to predict whether an individual's annual income exceeds \$50,000 (positive class) or not (negative class). This type of model is used in sensitive contexts (credit scoring, recruitment). The objective of this project is to conduct a complete audit to determine whether the model makes fair and transparent decisions, and to identify potential algorithmic discrimination risks.

## 2. Audit Scope

**Axis 1 — Data Quality.**

- *Analysis:* study of the dataset distribution (32,561 rows); identification of missing values (e.g., `workclass`, `occupation`) and handling strategies.
- *Sensitive attributes:* primary focus on `sex` (Male/Female); secondary focus on `race` (White/Non-White).
- *Initial finding to verify:* the dataset is historically imbalanced (males $\approx 67\%$ of the data and exhibit a much higher rate of incomes above \$50k).

**Axis 2 — Bias Measurement and Fairness.**

- *Target model:* baseline logistic regression or random forest.
- *Fairness metrics (via `Fairlearn`):*
  - *Demographic Parity Difference:* compare positive prediction rates between males and females.
  - *Equalized Odds:* compare True Positive Rate (TPR) and False Positive Rate (FPR) across groups.
- *Objective:* determine whether the model is significantly less performant for the minority group (females).

**Axis 3 — Explainability (XAI).**

- *Global (SHAP):* identify the most influential variables in the model's global predictions (e.g., `capital-gain`, `education-num`, `marital-status`).
- *Local (LIME):* select a borderline case to generate a textual explanation and check whether "proxy" variables (e.g., marital status *Married*, correlated with sex) unduly influence the decision.

**Axis 4 — Robustness.**

- *Test:* apply slight perturbations to the test data (e.g., noise on age or education) to check whether predictions flip aberrantly.
- *Objective:* ensure the model is not fragile under minor variations in candidate profiles.

## 3. Methodology and Tools

**Environment:** Python (Jupyter Notebook).
**Libraries:** `pandas` / `scikit-learn` (preparation and modeling), `Fairlearn` (fairness metrics), `SHAP` and `LIME` (explainability).
**Reproducibility:** fixed random seeds so that the notebook yields identical results on each run.

## 4. Team Organization and Planning

**Abada Zine Eddine Ahmed:** data preparation, model training, fairness metrics with `Fairlearn`.
**Naib Raja:** implementation of SHAP and LIME, robustness testing.
**Joint work:** writing the final audit report (findings, risk levels, recommendations), including a mitigation proposal (e.g., *Reweighing* from AIF360).